

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 7: Ratio and Proportion

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Properties of proportion

1. Use the properties of proportion to solve $\frac{6x^2 + 3x - 5}{3x - 5} = \frac{9x^2 + 2x + 5}{2x + 5}$, where $x \neq 0$. **ICSE 2025, 3 marks**

SOLUTION

Apply dividendo: if $\frac{a}{b} = \frac{c}{d}$ then $\frac{a-b}{b} = \frac{c-d}{d}$.

$$\frac{6x^2 + 3x - 5 - (3x - 5)}{3x - 5} = \frac{9x^2 + 2x + 5 - (2x + 5)}{2x + 5}, \text{ so } \frac{6x^2}{3x - 5} = \frac{9x^2}{2x + 5}$$

Since $x \neq 0$, divide both sides by $3x^2$: $\frac{2}{3x - 5} = \frac{3}{2x + 5}$

$$4x + 10 = 9x - 15, \text{ so } 5x = 25.$$

Answer: $x = 5$

2. Given $\frac{(a+b)^3}{(a-b)^3} = \frac{64}{27}$, find (a) $\frac{a+b}{a-b}$ (b) the ratio $a : b$, using the properties of proportion.

ICSE 2024, 4 marks

SOLUTION

(a) Take the cube root of both sides: $\frac{a+b}{a-b} = \frac{4}{3}$

(b) Apply componendo and dividendo: $\frac{(a+b) + (a-b)}{(a+b) - (a-b)} = \frac{4+3}{4-3}$

$$\frac{2a}{2b} = \frac{7}{1}$$

Answer: (a) $\frac{4}{3}$ (b) $a : b = 7 : 1$

3. Use componendo and dividendo to solve $\frac{\sqrt{2x+2} + \sqrt{2x-1}}{\sqrt{2x+2} - \sqrt{2x-1}} = 3$. **ICSE 2023, 3 marks**

SOLUTION

Write 3 as $\frac{3}{1}$ and apply componendo and dividendo: $\frac{2\sqrt{2x+2}}{2\sqrt{2x-1}} = \frac{3+1}{3-1} = 2$

Square both sides: $\frac{2x+2}{2x-1} = 4$, so $2x+2 = 8x-4$

$$6x = 6$$

Answer: $x = 1$

4. Find $x : y$, given $\frac{x^2 + 2x}{2x + 4} = \frac{y^2 + 3y}{3y + 9}$. **ICSE 2020, 3 marks**

SOLUTION

Factorise: $\frac{x(x + 2)}{2(x + 2)} = \frac{y(y + 3)}{3(y + 3)}$

$\frac{x}{2} = \frac{y}{3}$. By alternendo, $\frac{x}{y} = \frac{2}{3}$.

Answer: $x : y = 2 : 3$

5. If $x = \frac{\sqrt{2a+1} + \sqrt{2a-1}}{\sqrt{2a+1} - \sqrt{2a-1}}$, prove that $x^2 - 4ax + 1 = 0$. **ICSE 2020, 4 marks**

SOLUTION

Write x as $\frac{x+1}{x-1}$ and apply componendo and dividendo: $\frac{x+1}{x-1} = \frac{2\sqrt{2a+1}}{2\sqrt{2a-1}}$

Square both sides: $\frac{x^2 + 2x + 1}{x^2 - 2x + 1} = \frac{2a+1}{2a-1}$

Apply componendo and dividendo again: $\frac{2x^2 + 2}{4x} = \frac{4a}{2}$, so $\frac{x^2 + 1}{2x} = 2a$

$x^2 + 1 = 4ax$, that is $x^2 - 4ax + 1 = 0$. **Proved.**

6. Use the properties of proportion to solve $\frac{\sqrt{5x} + \sqrt{2x-6}}{\sqrt{5x} - \sqrt{2x-6}} = 4$. **ICSE 2019, 3 marks**

SOLUTION

Apply componendo and dividendo: $\frac{2\sqrt{5x}}{2\sqrt{2x-6}} = \frac{4+1}{4-1} = \frac{5}{3}$

Square both sides: $\frac{5x}{2x-6} = \frac{25}{9}$, so $45x = 50x - 150$

Answer: $x = 30$

7. Use the properties of proportion to solve $\frac{2x + \sqrt{4x^2 - 1}}{2x - \sqrt{4x^2 - 1}} = 4$, given that x is positive. **ICSE 2018, 3 marks**

SOLUTION

Apply componendo and dividendo: $\frac{4x}{2\sqrt{4x^2 - 1}} = \frac{4+1}{4-1}$, so $\frac{2x}{\sqrt{4x^2 - 1}} = \frac{5}{3}$

Square both sides: $\frac{4x^2}{4x^2 - 1} = \frac{25}{9}$, so $36x^2 = 100x^2 - 25$

$64x^2 = 25$, and x is positive.

Answer: $x = \frac{5}{8}$

8. Given $\frac{7m+2n}{7m-2n} = \frac{5}{3}$, use the properties of proportion to find (a) $m : n$ (b) $\frac{m^2+n^2}{m^2-n^2}$. **ICSE 2017, 3 marks**

SOLUTION

(a) Apply componendo and dividendo: $\frac{14m}{4n} = \frac{5+3}{5-3} = 4$, so $\frac{m}{n} = \frac{8}{7}$.

(b) $\frac{m^2}{n^2} = \frac{64}{49}$. Apply componendo and dividendo: $\frac{m^2+n^2}{m^2-n^2} = \frac{64+49}{64-49}$

Answer: (a) $8 : 7$ (b) $\frac{113}{15}$

2 Proportion

9. A train moves at a uniform speed. It covers 60 m in 2 s, 90 m in x s and y m in 5 s. Find x and y . (a) $x = 4, y = 150$ (b) $x = 3, y = 100$ (c) $x = 4, y = 100$ (d) $x = 3, y = 150$ **ICSE 2024, 1 mark**

SOLUTION

Speed = $\frac{60}{2} = 30$ m/s. At uniform speed, distance is proportional to time.

$x = \frac{90}{30} = 3$ and $y = 30 \times 5 = 150$

Answer: (d)

10. What number must be added to each of 4, 6, 8 and 11 so that the four numbers are in proportion? **ICSE 2023, 3 marks**

SOLUTION

Let the number be x . Then $\frac{4+x}{6+x} = \frac{8+x}{11+x}$.

$(4+x)(11+x) = (6+x)(8+x)$, so $44 + 15x + x^2 = 48 + 14x + x^2$

$x = 4$. Check: 8, 10, 12, 15 gives $8 \times 15 = 10 \times 12 = 120$.

Answer: 4

11. The numbers $K+3, K+2, 3K-7$ and $2K-3$ are in proportion. Find K . **ICSE 2019, 3 marks**

SOLUTION

$\frac{K+3}{K+2} = \frac{3K-7}{2K-3}$, so $(K+3)(2K-3) = (K+2)(3K-7)$

$2K^2 + 3K - 9 = 3K^2 - K - 14$, so $K^2 - 4K - 5 = 0$

$(K-5)(K+1) = 0$

Answer: $K = 5$ or $K = -1$

3 Continued proportion

12. Find the mean proportional between 4 and 9. (a) 4 (b) 6 (c) 9 (d) 36 **ICSE 2023, 1 mark**

SOLUTION

$$\text{Mean proportional} = \sqrt{4 \times 9} = \sqrt{36} = 6$$

Answer: (b)

13. b is the mean proportional between a and c . Show that $\frac{a^4 + a^2b^2 + b^4}{b^4 + b^2c^2 + c^4} = \frac{a^2}{c^2}$. **ICSE 2017, 3 marks**

SOLUTION

b is the mean proportional, so $b^2 = ac$.

$$\text{Numerator} = a^4 + a^2(ac) + (ac)^2 = a^2(a^2 + ac + c^2)$$

$$\text{Denominator} = (ac)^2 + (ac)c^2 + c^4 = c^2(a^2 + ac + c^2)$$

$$\text{LHS} = \frac{a^2(a^2 + ac + c^2)}{c^2(a^2 + ac + c^2)} = \frac{a^2}{c^2} = \text{RHS. Proved.}$$

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